CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

Chapter 7: Proportional Reasoning – 1

Topic-wise short notes with solved examples

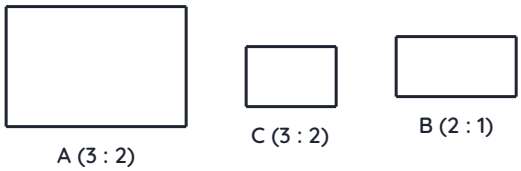
Ratios, proportion, the Rule of Three and cross multiplication, dividing a quantity in a given ratio, and using unit conversions in proportion problems.

1 Ratio

- A **ratio** $a : b$ compares two quantities by multiplication: for every a units of the first, there are b units of the second. The numbers a and b are the **terms**.
- Two pictures look alike (not stretched or squeezed) when their widths and heights change by the **same factor**, not by the same difference.
- **Simplest form**: divide both terms by their HCF. $60 : 40 = 3 : 2$.
- Both terms must be in the **same unit** before the ratio is written.
- Multiplying or dividing both terms by the same non-zero number does not change a ratio. Adding or subtracting the same number **does** change it.

Picture	Width (mm)	Height (mm)	Width : Height
A	60	40	$3 : 2$
B	40	20	$2 : 1$
C	30	20	$3 : 2$
D	90	60	$3 : 2$

- A, C and D have the same shape. B does not, although its width and height are each 20 mm less than those of A.



A and C have the same shape, but B is stretched

Example 1.1 Write the ratio of 600 mL to 1.5 L in simplest form.

SOLUTION

1.5 L $= 1500$ mL, so the ratio is $600 : 1500$.

HCF $= 300$

$2 : 5$

Example 1.2 A girl is 3 years old and her mother is 30. What is the ratio of their ages now, and after 9 years?

SOLUTION

Now: $3 : 30 = 1 : 10$. After 9 years: $12 : 39 = 4 : 13$.

The ratio changes, because the same number was added to both terms.

2 Proportion

- Two ratios are **in proportion** when they are equal in simplest form. We write $a : b :: c : d$.
- **Cross multiplication**: $a : b :: c : d$ exactly when $ad = bc$.
- To check a proportion, either reduce both ratios or cross multiply.

Example 2.1 Are $3 : 4$ and $72 : 96$ in proportion?

SOLUTION

HCF of 72 and 96 is 24, so $72 : 96 = 3 : 4$. Or check: $3 \times 96 = 288 = 4 \times 72$.

Yes

Example 2.2 Is $8 : 3 :: 24 : 6$ true?

SOLUTION

$8 \times 6 = 48$ but $3 \times 24 = 72$

No

Example 2.3 Fill in the blanks so that each ratio is proportional to $14 : 21$: $\square : 42$, $6 : \square$.

SOLUTION

$14 : 21 = 2 : 3$. For the first, $42 = 3 \times 14$, so the first term is $2 \times 14 = 28$.

For the second, $6 = 2 \times 3$, so the second term is $3 \times 3 = 9$.

$28 : 42$ **and** $6 : 9$

3 The Rule of Three

- When four quantities are in proportion and three are known, the fourth can be found. If $a : b :: c : d$ then

$$d = \frac{b \times c}{a}$$

- Āryabhaṭa (499 CE) called these **Rule of Three** (trairāśika) problems: multiply the "fruit" by the "requisition" and divide by the "measure".
- Another way: find the **factor of change** from a to c and apply it to b.
- Make the units match before setting up the proportion.

Example 3.1 For 120 students a cook uses 15 kg of rice. How much is needed for 80 students?

SOLUTION

$120 : 15 :: 80 : x$, so $x = \frac{15 \times 80}{120}$

10 kg

Example 3.2 A car covers 90 km in 150 minutes. At the same speed, how far does it go in 4 hours?

SOLUTION

4 hours $= 240$ minutes. $150 : 90 :: 240 : x$

$x = \frac{90 \times 240}{150}$

144 km

Example 3.3 6 glasses of lemonade need 10 spoons of sugar. How many spoons do 18 glasses need for the same sweetness?

SOLUTION

The factor of change is $18 \div 6 = 3$.

$10 \times 3 = 30$ **spoons**

Example 3.4 A tap fills a 500 mL mug in 15 seconds. How long does it take to fill a 10-litre bucket?

SOLUTION

10 L $= 10000$ mL. $500 : 15 :: 10000 : x$, so $x = \frac{15 \times 10000}{500} = 300$ seconds

5 minutes

4 When the Rule of Three does not apply

- The rule works only when the two quantities increase or decrease **together by the same factor**.

- Speed and time for a fixed distance do not behave this way: a higher speed means **less** time. Such situations are not direct proportions.
- Prices are often not proportional to pack size. A 6 mL sachet for ₹2 and a 180 mL bottle for ₹154 give $6 : 180 = 1 : 30$ for volume but $2 : 154 = 1 : 77$ for price.

Example 4.1 200 g of one tea costs ₹200. 1 kg of another tea costs ₹800. Are weight and price in proportion? Which tea costs more?

SOLUTION

In grams: $200 : 200 = 1 : 1$ and $1000 : 800 = 5 : 4$. Not in proportion.

Price of 1 kg of the first tea $= 5 \times 200$, that is ₹1000.

The first tea is more expensive (₹1000 against ₹800 per kg)

5 Comparing by unit rate

- To compare two situations, find the value **for one unit** in each.

Example 5.1 City P has 30 million people in 1500 sq km. City Q has 20 million people in 500 sq km. Which is more crowded?

SOLUTION

P: $\dfrac{30{,}000{,}000}{1500} = 20{,}000$ people per sq km. Q: $\dfrac{20{,}000{,}000}{500} = 40{,}000$ people per sq km.

City Q

Example 5.2 A 60 ft wall uses 3 bags of cement and a 40 ft wall uses 2 bags. Are the walls equally strong?

SOLUTION

$60 : 3 = 20 : 1$ and $40 : 2 = 20 : 1$

Yes. Each bag covers 20 ft in both walls.

6 Dividing in a given ratio

- To divide a quantity x in the ratio $m : n$, think of $m + n$ equal groups, each of size $\dfrac{x}{m+n}$.

$$\text{First part} = \frac{m}{m+n} \times x, \qquad \text{Second part} = \frac{n}{m+n} \times x$$

- The same idea works for three parts in the ratio $l : m : n$, using $l + m + n$ groups.

Example 6.1 Divide ₹4500 in the ratio $2 : 3$.

SOLUTION

$2 + 3 = 5$ groups, each of ₹900.

₹1800 and ₹2700

Example 6.2 Two partners invest ₹75,000 and ₹25,000 and share a profit of ₹4000 in the ratio of their investments. Find each share.

SOLUTION

$75000 : 25000 = 3 : 1$. One group $= 4000 \div 4$, that is ₹1000.

₹3000 and ₹1000

Example 6.3 Acid and water are mixed in the ratio $1 : 5$. How much of each is in 240 mL of the solution?

SOLUTION

One group $= 240 \div 6 = 40$ mL

40 mL of acid and 200 mL of water

7 Changing a mixture

- When something is added to a mixture, find the actual quantities first. The part that is not added stays the same.

Example 7.1 A 40 kg mixture has sand and cement in the ratio $3 : 1$. How much cement must be added to make the ratio $5 : 2$?

SOLUTION

Sand $= \frac{3}{4} \times 40 = 30$ kg and cement $= 10$ kg.

Sand stays 30 kg. $5 : 2 :: 30 : x$ gives $x = 12$ kg of cement.

Add 2 kg of cement

Example 7.2 Blue and yellow paint are mixed in the ratio $3 : 5$ to make 40 mL of green. Then 20 mL of yellow is added. Find the new ratio.

SOLUTION

Blue $= 15$ mL, yellow $= 25$ mL. After adding: yellow $= 45$ mL.

$15 : 45 = 1 : 3$

Example 7.3 A child is 1 year old and her brother is 5. When will the ratio of their ages be $1:2$?

SOLUTION

After x years: $\frac{1+x}{5+x}=\frac{1}{2}$, so $2+2x=5+x$ and $x=3$.

When she is 4 years old

8 Unit conversions

Quantity	Conversion
Length	1 metre $=3.281$ feet
Area	1 sq metre $=10.764$ sq feet
Area	1 acre $=43{,}560$ sq feet
Area	1 hectare $=10{,}000$ sq metres $=2.471$ acres
Volume	1 mL $=1$ cc; 1 litre $=1000$ mL
Temperature	$F=\frac{9}{5}C+32$ and $C=\frac{5}{9}(F-32)$

Example 8.1 Convert $25°$C to Fahrenheit.

SOLUTION

$F=\frac{9}{5}\times 25+32=45+32$

77°F

Example 8.2 One acre of land costs ₹15,00,000. Find the cost of 2400 sq ft.

SOLUTION

$43560:1500000::2400:x$, so $x=\frac{1500000\times 2400}{43560}$

About ₹82,645

9 Common mistakes

- Writing a ratio with different units, such as $1:800$ for 1 kg and ₹800 when the other ratio uses grams.
- Thinking equal differences keep a ratio the same. Only equal **factors** do.
- Reversing the order of terms: $3:5$ is not $5:3$.
- Dividing by m or n instead of $m+n$ when sharing in the ratio $m:n$.
- Using the Rule of Three for speed and time, or workers and days, where one quantity rises as the other falls.